AIR INDIA

CDO BRIEFING · TEAM 3 · IE MBDS · DATA: FEB–MAR 2022

A NEW AIR INDIA · THE OWNER'S OPENING DIAGNOSTIC

TWO LEVERS.
ONE DECISION.

300,153 flights. Zero fuel data — so we engineered it from physics. This is what optimising **fuel and price** actually looks like.

WHY THIS MATTERS NOW

A national carrier, just privatised.

On **27 January 2022**, Tata acquired 100% of Air India for **₹18,000 crore** — ending 69 years of state ownership. This data was scraped in the weeks that followed. It is, literally, the first thing the new owner should have looked at.

MARKET LEADER (INDIGO)

55.7%

Air India + Vistara combined: 18.4% — the clear #2 bloc that later merged.

INDIA DOMESTIC PAX, 2022

110M

+52% year on year. A market rebounding hard post-COVID.

JET FUEL ≈ SHARE OF COST

40%

And ATF prices spiked through 2022 on the Ukraine shock. Fuel is the #1 controllable cost.

Sources: Tata Group · DGCA · IndianOil ATF · IATA/CAPA

THE BRIEF, AND ITS TRAP

We were asked to optimize fuel & price. The data has no fuel.

The dataset records price, route, timing and stops — but not a single litre of fuel.
Most teams will quietly skip the fuel half of the question. We refused to.

So we engineered a fuel model from physics — great-circle distance between cities plus a takeoff-cycle penalty per stop, calibrated to real A320 burn rates and Feb–Mar 2022 jet-fuel prices.

That lets us answer the half of the brief the raw data cannot — and put fuel and revenue on the **same chart**. Every fuel figure is a labelled estimate, not invented data.

THE PICTURE IN FOUR NUMBERS

What 300,153 flights revealed.

PANIC-TAX MULTIPLE	BUSINESS-CLASS PREMIUM	FLIGHTS ANALYSED	UNDER-PRICED BACKLOG
2.8×	8×	300K	62.6%
₹14,226 last-minute vs ₹5,037 booked early. Same seat, same cost.	Sold by only 2 of 6 carriers — both are Tata's.	30 routes · 6 airlines · India's six metros.	of economy seats sell in the cheapest booking window.

Three of these are money left on the runway. One is a moat. We'll take each in turn.

THE HERO VIEW · FUEL AND PRICE ARE ONE DECISION

The Fuel–Revenue Efficiency Frontier

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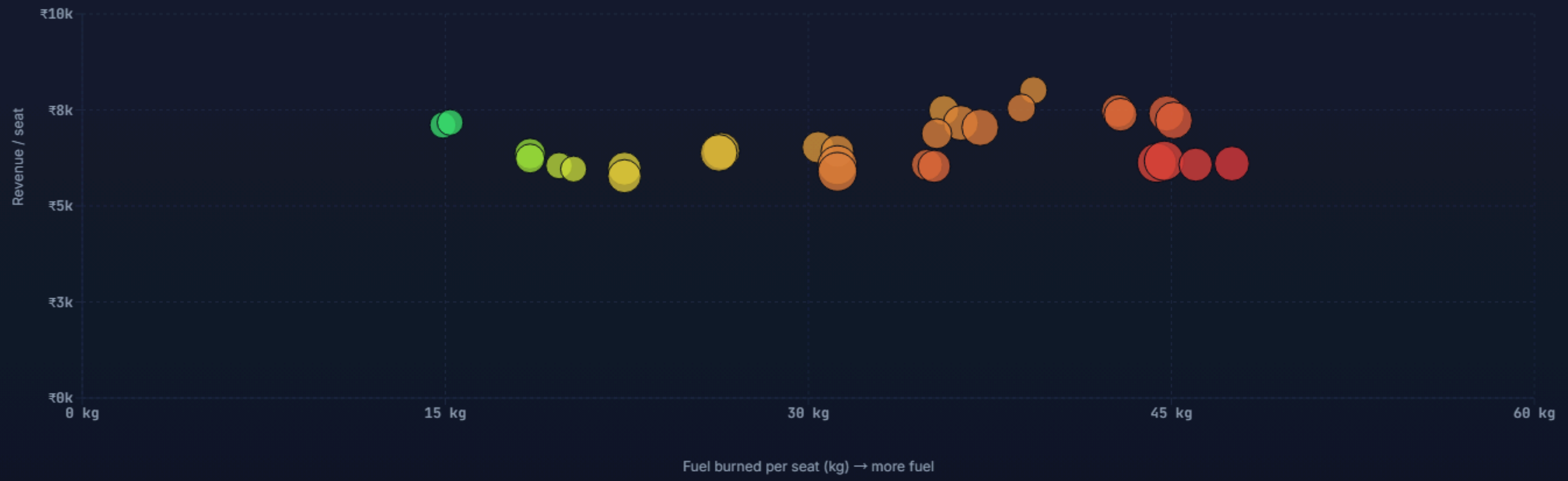
Every route: modeled fuel burned per seat (x) against revenue per seat (y). Bubble size = booking volume. Green earns the most revenue per kilogram of fuel; red earns the least.



Most efficient: Bangalore → Chennai earns ₹7,106 on just 14.9 kg/seat. Least efficient: Delhi → Chennai earns less (₹6,102) on 47.5 kg — 3.7× worse revenue-per-fuel, priced almost identically. Price does not reflect fuel cost.

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FINDING 01 · REVENUE

The Panic Tax

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Average economy fare by days before departure. The fare is flat and cheap until ~20 days out, then climbs steeply as seats run low.

Book at T-1 and you pay **2.8×** the far-out fare — yet **62.6%** of seats sell in that cheapest window. The blue line marks where the cliff begins (~20 days).

- **We are giving seats away early, then gouging late.** The same seat swings 2.8× in price with zero change in cost. That's not pricing — it's leaving money on the runway in both directions.

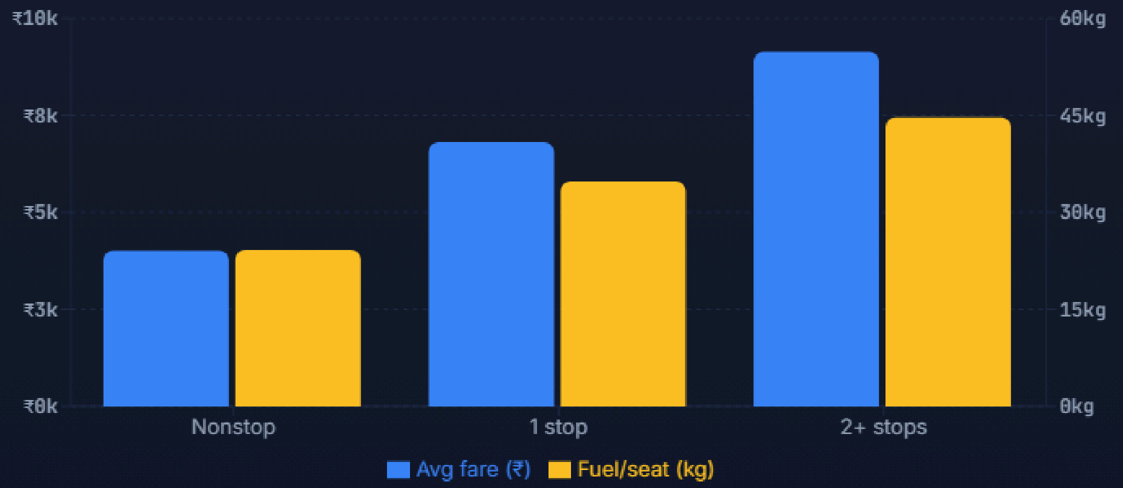
A disciplined dynamic-pricing curve lifts the floor on early bookings — where **62.6%** of demand sits — without needing the punishing last-minute spike that erodes trust.

- We **give seats away early, then gouge late** — a 2.8× swing with zero change in cost. Yet 62.6% of demand sits in the cheap early window we under-price. That is money lost in both directions.

Connections cost twice

The Double-Crime of Connections

More stops charge the passenger more **and** burn more fuel per seat — every extra takeoff is the thirstiest phase of flight. Cost and price move together.



Nonstop is cheapest for the passenger and lightest on fuel. Two-plus-stop itineraries are the worst on both — a rare case where the customer-friendly move is also the green move.

- **The rare win-win-win.** Cutting wasteful connections lowers the fare, the fuel bill, and the carbon footprint at the same time. Customer, CFO, and ESG officer all cheer.

Takeoff and climb are the thirstiest phases of any flight. Every avoided stop is a takeoff cycle of fuel saved — which is why our model penalises stops directly rather than trusting gate-to-gate duration.

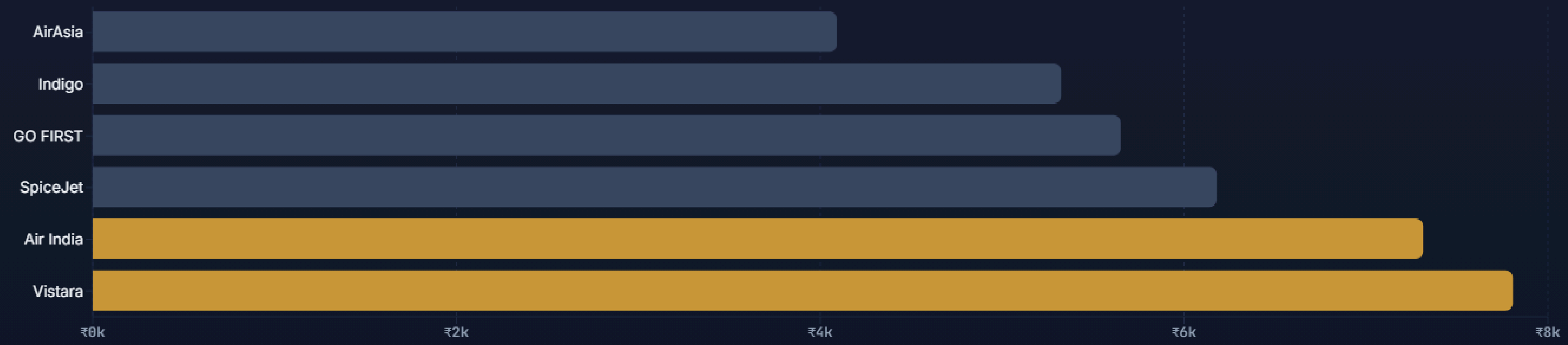
- A rare **win-win-win**: 2+ stop trips charge passengers more (₹9,142 vs ₹4,013) *and* burn 45 kg/seat vs 24. Cutting them lowers fare, fuel bill, and carbon at once.

FINDING 03 · PREMIUM

The whitespace only Tata can hold

The Premium Whitespace

Business class commands a **8× premium** (₹52,540 vs ₹6,572) — yet only **2 of 6** carriers sell it. Gold bars below already monetise the premium cabin; grey bars leave it on the table.



Air India and Vistara — the two Tata carriers — are the **only** two selling business class, at ₹47,131 and ₹55,477 respectively. The merged group owns India's premium-cabin supply outright.

- Air India and Vistara are the **only two carriers selling business class**, at an 8× premium. The merged group owns India's premium-cabin supply outright — a moat low-cost rivals can't cross overnight.

WHAT'S AT STAKE

Risks & opportunities.

Risks of standing still

- **Revenue leakage** — the early-booking discount is a structural giveaway, repeated on every flight, every day.
- **Fuel & ESG exposure** — inefficient routings burn cash and carbon as ATF prices climb and emissions reporting tightens.
- **Premium erosion** — if a low-cost rival adds business class, the moat narrows before integration even completes.

Opportunities to seize

- **Smarter pricing curve** — lift the floor on the 62.6% of seats sold early, without the punishing last-minute spike.
- **Network re-routing** — move 2+ stop traffic onto nonstop/1-stop metal for a triple win.
- **Premium consolidation** — make the merged group's business cabin the default choice for corporate India.

THE DECISION

Three moves for day one.

MOVE 01

Re-time the revenue curve

Deploy disciplined dynamic pricing that lifts early fares modestly instead of giving the seat away. 129,271 far-out bookings sit ₹9,189 below the late fare.

+ ₹1,378 / seat at a conservative 15% capture

MOVE 02

Kill the inefficient connections

Re-route 2+ stop itineraries onto nonstop and single-connection metal. Lower fare for the passenger, lower fuel bill for us.

- 9.9 kg fuel & 31.3 kg CO_2 / seat

MOVE 03

Own the premium cabin

Protect and grow business class as the merged group's profit engine before low-cost rivals copy it. It already commands an 8× premium.

8× revenue per premium seat vs economy

WHAT HAPPENS NEXT

Act, or leave it on the runway.

If we act

- ▲ Capture a structural pricing uplift on the majority of seats sold — every route, every day.
- ▲ Cut fuel and CO_2 per seat on the worst-routed traffic, hedging the ATF spike.
- ▲ Lock in the premium moat before integration with Vistara completes.
- ▲ Turn the first data diagnostic into a repeatable revenue-management capability.

If we don’t

- ▼ The early-booking giveaway compounds across 110M+ passengers a year.
- ▼ Fuel waste scales with the post-COVID traffic boom — against us.
- ▼ A rival adds business class and the 8× premium starts to erode.
- ▼ The turnaround story stalls on the metrics the market watches.

THE ASK

Fuel and price are not two problems. They are one decision.

Approve a 90-day pilot: dynamic-pricing floor on early bookings, a re-routing review of multi-stop traffic, and a premium-cabin growth plan. We have the engine. We have the data. We know exactly what to do.

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APPENDIX · FOR THE Q&A

Why this is real, and how we built it.

The data

300,153 real flights scraped from EaseMyTrip across India's six metros, Feb–Mar 2022. Zero missing values, zero duplicates. Internally coherent in ways synthetic data isn't — e.g. business class sold only by the two full-service carriers, exactly matching the real 2022 market.

The fuel model

Estimated from first principles, every constant a published benchmark:

Great-circle distance (6 metros)	haversine
Cruise burn (A320 family)	2,400 kg/hr
Takeoff-cycle penalty / stop	825 kg
CO_2 factor (ICAO)	3.16×
ATF price (Delhi, Feb–Mar '22)	₹100 / L

Fuel, cost and CO_2 are **modeled estimates**, labelled as such throughout. Everything runs live in a full-stack app — ask any route, the model computes it on the spot.